
WBI11 — Rapid Revision

IAL Biology Unit 1: Molecules, Diet, Transport and Health

Last-minute cram sheets — designed to be scanned, not read

Topic priority ordered by frequency across 22 past papers (Jan 2019 – May 2026)

80 marks · 90 minutes · 8 questions

Q1–Q4 ≈ 29 marks | Q5–Q8 ≈ 50 marks

TRIAGE — what to review, in what order

Time left	Cover these	Why
30 min	Sheets 1, 3, 5, 6 — carbohydrates, enzymes, blood vessels, atherosclerosis + clotting	Appear in 16-20 of 22 papers. Highest marks per minute of revision.
2 hours	Add sheets 2, 4, 7, 9 — lipids, proteins, cardiac cycle, membranes + transport	Appear in 14-18 of 22 papers.
Half a day	Add sheets 8, 10, 11, 12 — CF, DNA and protein synthesis, inheritance, screening	7-13 of 22, but heavy marks when they appear.
Always	Sheet 13 (calculations) and sheet 14 (phrase triggers)	Calculations are in every paper. Half the paper is explanation marks.

THE FOUR RULES, IF YOU REMEMBER NOTHING ELSE
1. "Explain" means give a reason — your sentence needs *because*, *so* or *therefore*. **2.** Quote figures **with units** from every graph and table. **3.** Show every line of every calculation. **4.** Don't over-write on Q1-Q4; Q5-Q8 hold 62% of the marks.

SHEET 1 — WATER & CARBOHYDRATES (20/22 papers)

WATER — DIPOLAR, HYDROGEN BONDS EXPLAIN EVERYTHING

Property	Importance
Solvent (polar)	Transport in blood/cytoplasm; reactions occur in solution
High specific heat capacity	Buffers temperature change
High latent heat of vaporisation	Efficient cooling by sweating, little water lost
Cohesion / surface tension	Continuous water columns; surface habitat
Ice less dense (open lattice)	Insulates water below; aquatic life survives

CARBOHYDRATES

- **Monosaccharides** glucose, fructose, galactose · formula (CH₂O)_n · **α-glucose** OH below C1, **β-glucose** OH above C1
- **Condensation** → **glycosidic bond** + **1 water released**. **Hydrolysis** = reverse, **uses** water.
- **Maltose** = glu+glu · **Sucrose** = glu+fructose · **Lactose** = glu+galactose

	Structure	Function link
Amylose	α-glucose, 1,4 only , unbranched, helix	Compact → lots stored in small space
Amylopectin	α-glucose, 1,4 + 1,6 , branched	Many ends → rapid hydrolysis
Glycogen	α-glucose, 1,4 + 1,6, more branched still	Fastest release; animals, liver + muscle

MUST SAY — WHY POLYSACCHARIDES SUIT STORAGE (4 MARKS)
Insoluble → **no osmotic effect** · **compact** → much stored in small volume · **branched** → many ends for **rapid hydrolysis** · **large** → cannot diffuse out of cell

SHEET 2 — LIPIDS & PROTEINS (15-17/22 papers)

LIPIDS

- **Triglyceride** = 1 glycerol + **3 fatty acids, ester bonds**, condensation, **3 waters released**
- **Saturated**: no C=C · straight chains · **pack closely** · solid at room temp
- **Unsaturated**: one or more C=C → **kinks** → **pack less closely** → liquid (oils). Mono = 1 double bond, poly = more than 1
- More energy per gram than carbohydrate — **more C-H bonds, less oxygen** (more reduced)
- **Phospholipid**: hydrophilic phosphate head + 2 hydrophobic tails → forms bilayer

PROTEINS

Amino acid = central C + -NH₂ + -COOH + H + **R group**. Join by condensation → **peptide bond**.

Level	What	Bonds
Primary	Sequence of amino acids	Peptide
Secondary	α-helix / β-pleated sheet	H bonds between C=O and N-H of backbone
Tertiary	3D folding of one chain	H bonds, ionic, disulfide bridges , hydrophobic — all between R groups
Quaternary	>1 polypeptide chain	Same types, between chains

	Globular	Fibrous
Shape / solubility	Compact, spherical, soluble (hydrophilic R groups outward)	Long parallel chains, insoluble
Role	Metabolic — enzymes, haemoglobin, antibodies	Structural — collagen, keratin

Collagen: 3 chains in a **triple helix** · H bonds · **glycine every 3rd amino acid** (small enough for the centre) · **cross-links** + **staggered ends** → tensile strength

SHEET 3 — ENZYMES (16/22 papers · "active site" = most frequent MS term)

- Globular proteins · biological catalysts · **lower the activation energy**
- **Active site complementary** to substrate → **enzyme-substrate complex**
- **Lock and key** = exact fit already · **Induced fit** = active site **changes shape slightly** on binding, straining substrate bonds (the accepted model)

Factor	Explanation that scores
Temperature ↑	More kinetic energy → more frequent successful collisions → more E-S complexes. Above optimum: H and ionic bonds holding tertiary structure break → active site changes shape → no longer complementary → denatured
pH	R groups change charge → ionic and H bonds disrupted → active site shape altered
[Substrate] ↑	Rate rises until all active sites occupied (saturated) → plateau at V _{max} ; enzyme concentration now limiting
Competitive inhibitor	Similar shape → binds active site → effect reduced by raising [substrate]
Non-competitive	Binds elsewhere → changes active site shape → not overcome by raising [substrate]

NEVER WRITE "the enzyme is killed / destroyed / melts". Denaturation breaks the bonds maintaining **tertiary** structure — **not** peptide bonds. Primary structure is unchanged.

SHEET 4 — HEART & CARDIAC CYCLE (8/22 papers, but heavy marks + a timing calculation)

- **Left ventricle thicker** → pumps to **whole body** at higher pressure; right pumps only to lungs at lower pressure (protects alveolar capillaries)
- **AV valves** = tricuspid (R), bicuspid/mitral (L) → stop backflow into atria. **Semilunar** = pulmonary + aortic → stop backflow into ventricles
- **Coronary arteries** supply heart muscle itself

Stage	Event	Valves
Atrial systole	Atria contract, push remaining blood into ventricles	AV open, SL closed
Ventricular systole	Ventricles contract, pressure rises above atria then above arteries	AV close (1st sound) → SL open
Diastole	All relax, pressure falls, blood flows in passively	SL close (2nd sound), AV open

THE VALVE MARK, EVERY TIME A valve opens because the **pressure behind it is greater than the pressure in front of it**.
Cardiac output = stroke volume × heart rate · heart rate = 60 ÷ cycle length in seconds

SHEET 5 — BLOOD VESSELS & TISSUE FLUID (18/22 papers)

Vessel	Structure	Function it serves
Artery	Thick wall, much elastic tissue + smooth muscle , narrow lumen, no valves	Withstands high pressure; elastic tissue stretches and recoils → maintains pressure, smooths pulsatile flow
Arteriole	More smooth muscle, less elastic tissue	Vasoconstriction / vasodilation redirects blood
Capillary	One cell thick squamous endothelium, pores, narrow lumen, huge total cross-sectional area	Short diffusion distance + large surface area ; large total area slows blood → more time for exchange
Vein	Thin wall, wide lumen , valves	Low pressure; wide lumen = less resistance; valves prevent backflow ; skeletal muscle squeezes blood on

TISSUE FLUID — THE TWO-END EXPLANATION **Arterial end: hydrostatic pressure** > opposing osmotic effect of plasma proteins → fluid **forced out**. Proteins and cells too large to leave.
Venous end: hydrostatic pressure has **fallen**; plasma proteins have **lowered the water potential** of blood → water **returns by osmosis**. Excess drains into the **lymphatic system**.

SHEET 6 — ATHEROSCLEROSIS, CLOTTING & RISK (16/22 papers)

ATHEROSCLEROSIS — 5 STEPS

1. Endothelium damaged (high BP, smoking toxins) → **2. inflammatory response**, white blood cells enter wall → **3.** accumulate **cholesterol and lipids** → **fatty streak** → **4.** builds into **atheroma/plaque**, hardened by fibrous tissue and calcium salts → **5. lumen narrows** → **blood pressure rises** → more damage = **positive feedback**.
Consequences: angina (narrowed coronary artery, less O₂ to heart muscle) · myocardial infarction (full blockage, cardiac muscle dies) · stroke (artery to brain blocked or ruptured) · aneurysm (weakened wall balloons, may rupture)

CLOTTING CASCADE — IN ORDER, 3 MARKS

1. Damage → platelets and damaged tissue release **thromboplastin**
2. Thromboplastin + **calcium ions** (and vitamin K) → **prothrombin** → **thrombin**
3. Thrombin (an enzyme) → **soluble fibrinogen** → **insoluble fibrin** → **mesh** traps platelets and red cells
Clot inside an intact vessel = **thrombus**; broken free and lodged elsewhere = **embolus**.

RISK AND CORRELATION

"DOES THIS PROVE X CAUSES Y?" — THE REQUIRED ANSWER It shows a **correlation**, which **does not prove causation**. Then give two or more: **confounding variables** not controlled · **sample size** too small or **unrepresentative** · no **control group** · population association, not individual mechanism · no indication of **statistical significance** · single time period/place/age group.

CVD risk factors. Non-modifiable: age, sex (male higher), genetics/family history. Modifiable: smoking, high blood pressure, high blood cholesterol, obesity, inactivity, diet high in saturated fat and salt, excess alcohol, diabetes.

SHEET 7 — CHOLESTEROL, DIET & TREATMENTS (18/22 and 11/22 papers)

	LDL	HDL
Carries	Cholesterol liver → tissues	Cholesterol tissues → liver for breakdown/excretion
How	Binds LDL receptors . High saturated fat saturates the receptors → excess LDL stays in blood → deposited in artery walls	Lowers blood cholesterol; can remove cholesterol from plaques
Risk	Increases	Reduces

BMI = mass (kg) ÷ height (m)² · waist-to-hip ratio for central obesity · energy budget = intake vs expenditure, surplus stored as fat · antioxidants (vitamin C) reduce free-radical damage

Drug	Mechanism	Side effects
Antihypertensives	Lower blood pressure → less endothelial damage, less workload on heart	Fatigue, dizziness, headaches, swollen ankles
Statins	Inhibit the enzyme catalysing cholesterol synthesis in the liver → lower LDL	Muscle pain and weakness, liver damage
Anticoagulants	Reduce clot formation	Uncontrolled bleeding; dose must be monitored
Platelet inhibitors	Reduce platelet aggregation	Bleeding, stomach irritation and ulcers

Evaluation questions: weigh risk reduction against side effects, note lifestyle change has none, and **state a judgement** — no judgement means no top band.

SHEET 8 — GAS EXCHANGE & FICK'S LAW (6/22 papers)

Rate of diffusion ∝ (surface area × concentration difference) ÷ diffusion distance

- Large surface area — many folded alveoli
- Short diffusion distance — alveolar wall and capillary endothelium each **one cell thick** (squamous)
- Steep gradient maintained — **ventilation** brings fresh air, **blood flow** carries oxygen away
- As size increases, **SA:V ratio falls** → diffusion alone too slow → specialised exchange surface + mass transport system needed

SHEET 9 — MEMBRANES & TRANSPORT (14/22 papers)

Fluid mosaic model: phospholipid bilayer, hydrophilic heads out, hydrophobic tails in. **Fluid** = phospholipids move laterally; **mosaic** = proteins scattered through. Plus **channel/carrier proteins**, **cholesterol** (regulates fluidity), **glycoproteins/glycolipids** (recognition, receptors).

Process	Energy	Direction	Key detail
Simple diffusion	Passive	Down gradient	Small/non-polar straight through bilayer — O ₂ , CO ₂
Facilitated diffusion	Passive	Down gradient	Via channel or carrier proteins — ions, glucose, amino acids
Osmosis	Passive	High → low water potential	Pure water Ψ = 0 ; adding solute makes it more negative
Active transport	ATP	Against gradient	Carrier protein changes shape using ATP
Endo/exocytosis	ATP	Bulk, either way	Vesicles fuse with or pinch off membrane

VERBATIM DEFINITIONS **Osmosis** — movement of water from a region of **higher water potential** to a region of **lower water potential** through a **partially permeable membrane**.
Active transport — movement **against** a concentration gradient using **carrier proteins** and energy from **ATP**.
Never define osmosis by "water concentration".

SHEET 10 — CYSTIC FIBROSIS (7/22 papers)

THE CHAIN — GENE TO SYMPTOM, ALWAYS REQUIRED IN FULL CFTR gene (chromosome 7) → codes for **CFTR protein**, a **chloride ion channel** in epithelial cell membranes. Commonest mutation = **deletion of 3 bases** → one amino acid (phenylalanine) missing → protein folds incorrectly.
→ **Cl⁻ not secreted** → **Na⁺ absorbed** from mucus into cell → **water follows by osmosis** out of mucus → mucus **thick and sticky**.

System	Effect	Consequence
Gas exchange	Mucus blocks bronchioles, hard for cilia to clear, traps bacteria	Reduced surface area, increased diffusion distance → less gas exchange; repeated infections; alveolar damage
Digestive	Pancreatic duct blocked → enzymes can't reach small intestine	Food not digested/absorbed → poor growth, weight loss; enzyme supplements with meals
Reproductive	Males: sperm duct blocked or absent. Females: thick cervical mucus	Reduced fertility in both sexes

SHEET 11 — DNA, REPLICATION & PROTEIN SYNTHESIS (11–13/22 papers)

- **Nucleotide** = deoxyribose + **phosphate** + **base**. Joined by **phosphodiester bonds**; two **antiparallel** strands in a **double helix**
- **A-T = 2 hydrogen bonds** · **C-G = 3 hydrogen bonds** · purine always pairs with pyrimidine → constant helix width

SEMI-CONSERVATIVE REPLICATION **DNA helicase** breaks the **hydrogen bonds** and unwinds → each strand acts as a **template** → free nucleotides align by **complementary base pairing** → **DNA polymerase** joins them by forming **phosphodiester bonds** → each new molecule has **one original and one new strand**.

Meselson & Stahl: ¹⁵N → ¹⁴N, density gradient centrifugation. After **1 generation all intermediate** → rules out **conservative**. After **2 generations, half intermediate half light** → rules out **dispersive** → supports semi-conservative.

Stage	Where	What happens
Transcription	Nucleus	Helicase unwinds, H bonds break · RNA polymerase moves along the template (antisense) strand joining complementary RNA nucleotides, uracil replaces thymine · introns spliced out · mRNA leaves via nuclear pore
Translation	Ribosome	mRNA binds ribosome · tRNA anticodon complementary to mRNA codon brings its amino acid · peptide bonds form · ribosome moves codon by codon to a stop codon

Genetic code: triplet · non-overlapping · **degenerate** (most amino acids have >1 codon) · near-universal. **Codon** = mRNA. **Anticodon** = tRNA. Don't swap them.

MUTATION

- **Substitution** — often **no effect**, because the code is **degenerate** and the new codon may specify the same amino acid. If it does change it: **primary structure** changes → **tertiary structure** changes → function/active site changes
- **Deletion/insertion** — **frameshift**: every subsequent codon read differently → most of the sequence after is wrong → usually far more damaging

SHEET 12 — INHERITANCE, SCREENING & GENE THERAPY (12/22 and 2/22 papers)

Allele version of a gene · **dominant** expressed in heterozygote · **recessive** expressed only in homozygote · **genotype** alleles present · **phenotype** observable · **carrier** unaffected heterozygote

GENETIC DIAGRAM — 5 THINGS THAT CARRY MARKS **1.** Key defining symbols **2.** Parental genotypes + phenotypes **3.** Gametes, circled **4.** Punnett square **5.** Phenotypes with the **ratio or probability** actually asked for
Two carriers → **1 in 4 (25%)** affected. An **unaffected** child of two carriers is a carrier with probability **2 in 3** — frequently asked, frequently wrong.

Pedigrees: two unaffected parents with an affected child → **recessive**. Roughly equal in both sexes → **autosomal**. If asked "to what extent does this prove it", acknowledge the evidence but note the **small sample** and that other patterns aren't fully excluded.

Technique	Method / timing	Evaluation
Amniocentesis	Needle into amniotic sac, 15-17 weeks , fetal cells cultured	Reliable but late ; small miscarriage risk
CVS	Placental tissue, 8-12 weeks	Earlier result; somewhat higher miscarriage risk
PGD	IVF embryos, one cell removed and tested, unaffected implanted	No termination decision, but expensive, invasive, embryo-selection concerns

Ethics bank: informed consent and the right not to know · confidentiality and discrimination by insurers/employers · the termination decision · risk of the procedure · psychological impact · selecting non-medical traits.

Gene therapy: **somatic** = body cells, **not inherited**, must be repeated · **germ line** = all cells, **heritable**, prohibited in most countries. For CF: normal CFTR allele in **liposomes** or a modified virus, delivered to lung epithelium by **aerosol**.

SHEET 13 — CALCULATIONS (in 22/22 papers, ~4 per paper)

Type	Method	Trap
Percentage change	change ÷ original × 100	Dividing by the final value
Ratio	Divide both by the smaller; simplify	Writing it in the wrong order
Rate	change ÷ time. For a curve: draw a tangent , take its gradient	Omitting units
Cardiac cycle	Read durations off the pressure-time graph. HR = 60 ÷ cycle length (s)	Reading the wrong pair of points
SA : V ratio	Compute both, then divide or express as a ratio	Forgetting there are 6 faces on a cube
BMI	mass (kg) ÷ height (m) ²	Not converting cm → m first
Water potential	Ψ = Ψ _s + Ψ _p	Losing a negative sign
Magnification	actual = image ÷ magnification	Not converting units before dividing
Risk / probability	Give it as the question asks — fraction, % or "1 in n"	Answering in the wrong form

FOUR RULES **1.** Show every step — method marks survive an arithmetic slip **2.** Units, and the s.f. or d.p. asked for **3.** Convert units **before** calculating **4.** Sanity-check the size of the answer

SHEET 14 — PHRASE TRIGGERS

Scan this last. Left column is what the question is about; right column is the exact wording that earns the mark.

If the question is about...	...the phrase they want is
Enzyme and substrate	Active site complementary to the substrate → enzyme-substrate complex
High temperature / extreme pH	H and ionic bonds in the tertiary structure break → active site changes shape → no longer complementary → denatured
Any exchange surface	Large surface area · short diffusion distance · steep concentration gradient maintained
Water movement	Higher to lower water potential through a partially permeable membrane
Moving against a gradient	Carrier protein and ATP
DNA strands held together	Complementary base pairing : A-T 2 hydrogen bonds , C-G 3
A graph showing an association	Correlation does not prove causation + confounding variables + sample size
A heart valve	Opens because pressure behind > pressure in front
Artery wall	Elastic tissue stretches and recoils → maintains pressure, smooths flow
Why capillaries exchange well	Large total cross-sectional area slows the blood → more time for exchange
Starch or glycogen as a store	Insoluble, no osmotic effect · compact · branched → rapid hydrolysis
Unsaturated fats	Double bonds cause kinks → chains pack less closely
Effect of a mutation	Primary → tertiary structure changes → active site / function changes
A substitution with no effect	The code is degenerate → new codon may specify the same amino acid
Describing any graph	Quote figures with units and name the trend — "rises from 2.1 to 6.8 between 10 and 40 min, then plateaus"

SHEET 15 — 60-SECOND SELF-TEST

Cover the right column. If you hesitate, go back to that sheet.

Question	Answer
Bond joining two monosaccharides, and what's released?	Glycosidic bond; one water molecule
Difference between amylose and amylopectin?	Amylose 1,4 only and unbranched; amylopectin 1,4 and 1,6 and branched
Why are unsaturated lipids liquid at room temperature?	C=C bonds cause kinks, so chains pack less closely
Bonds in secondary vs tertiary structure?	Secondary: H bonds in the backbone. Tertiary: H, ionic, disulfide, hydrophobic — between R groups
What does an enzyme do to activation energy?	Lowers it
How is a non-competitive inhibitor different?	Binds elsewhere, changes active site shape, not overcome by more substrate
Why is the left ventricle wall thicker?	Pumps to the whole body at higher pressure
What makes a heart valve open?	Pressure behind it exceeds pressure in front
Why does blood slow in capillaries?	Large total cross-sectional area — allows time for exchange
What returns tissue fluid at the venous end?	Osmosis — plasma proteins have lowered the blood's water potential
First step of atherosclerosis?	Damage to the endothelium
Thrombin converts what into what?	Soluble fibrinogen into insoluble fibrin
Which lipoprotein raises CVD risk, and why?	LDL — carries cholesterol to tissues; receptors saturate, LDL stays in blood
State Fick's law.	Rate ∝ (surface area × concentration difference) ÷ diffusion distance
Define osmosis.	Water from higher to lower water potential through a partially permeable membrane
Two processes needing ATP?	Active transport; endo/exocytosis
What does CFTR transport, and what follows?	Chloride ions; Na ⁺ then absorbed, water follows by osmosis, mucus thickens
Why do CF patients need enzyme supplements?	Mucus blocks the pancreatic duct, so digestive enzymes can't reach the small intestine
How many H bonds between C and G?	Three
Two enzymes in DNA replication?	DNA helicase; DNA polymerase
What did Meselson and Stahl's second generation rule out?	Dispersive replication
Which strand is transcribed?	The template (antisense) strand
Codon vs anticodon?	Codon on mRNA; anticodon on tRNA
Why is a frameshift worse than a substitution?	Every subsequent codon is read differently
Two carriers — probability of an affected child?	1 in 4 (25%)
Earliest of the three screening techniques?	PGD (before implantation); then CVS at 8–12 weeks
Why is germ line therapy prohibited in most countries?	The change is heritable, affecting all cells of the individual and their descendants
BMI formula?	mass (kg) ÷ height (m) ²

WALKING INTO THE EXAM Underline the command word. Count the marks, then count your points. Give a reason wherever "explain" appears. Quote figures with units. Show every calculation line. Leave Q1–Q4 behind on time — half the paper is waiting in Q5–Q8.